

Robotics Software Developer

JOSEF GSTOETTNER

ROS/ROS2 | C++ | Python | Gazebo

CONTACT

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https://github.com/JosefGst
https://josefgst.github.io/portfolio/
Linkedin

SKILLS

Areas of specialization

- ROS / ROS 2 (Navigation, SLAM, Moveit)
- Robot Simulation in Gazebo & Isaacsim
- Behavior trees, robot task management
- Embedded software (STM32, Arduino)
- Single board computers (Raspberry Pi, Jetson)

Programming/Frameworks

Python	6 yrs
C / C++	6 yrs
ROS / ROS 2	4 yrs
Docker	2 yrs
C#	1 yrs
Matlab	1 yrs
JavaScript	1 yr
HTML, CSS	1 yr

OS: Windows | Ubuntu

Languages

English	Full Professional
Mandarin	Limited Working
German	Native

Learning & Hobbies

PCB design with KiCAD
Android development

6 yrs C/C++ and Python | 4 yrs ROS 2, navigation, SLAM and behavior trees for flexible and reliable task execution on autonomous mobile robots.

Work History

Senior Robotics Software Engineer 2026/3–present
ROS 2 ROBOT PROGRAMMING · Primech Ai | Sciences Park
- Set up servo motor control, navigation & SLAM for mecanum-wheeled mobile manipulator.
- Package robot software and tools with Docker for convenient development.
- Implement Open-RMF to manage an autonomous fleet of toilet-cleaning robots.
- Setup CI/CD for automatic testing of the whole ROS2 software and building the docker image.
- Collect Image Data and train the Object Detection using Yolo.

Assistant Research Engineer 7 mos 2025/9–2026/3
ROS 2 ROBOT PROGRAMMING · HKCRC | Sciences Park
- Develop autonomous tasks and recovery behaviors for a water tank cleaning robot using behavior trees.
- Set up navigation, SLAM & coverage planner to fully clean the whole surface.
- Packaged robot software and tools with Docker for easy deployment.
- Create realistic Gazebo simulations for faster iterations and testing.

System Engineer 2 yr 5 mos 2022/4–2024/8
ROS 2 ROBOT PROGRAMMING · LSCM | Sciences Park
- Set up navigate & SLAM (Cartographer, SlamToolbox, ...) for 2D/3D mapping of large areas.
- Simulated robots in Gazebo for testing and faster development.
- Developed autonomous docking utilizing AprilTags for precise movement.
- Tuned PID controllers for smooth movement and speed control.

Research Assistant 1 yr 8 mos 2020/8–2022/3
EMBEDDED SOFTWARE · HKUST | Clear Water Bay
- CAD design for 3D printing and laser cutting of prototypes.
- Firmware development on a low-power IoT accelerometer with BLE Mesh for predictive maintenance based on Nrf52.

Mechanical Engineer 10 mos 2019/8–2020/4
CAD DESIGN · KALBAS | Cheung Sha Wan

Education

Master of Science: Mechanical Engineering 2018/9–2019/6
· HKUST | Clear Water Bay

One Year Mandarin language study 2017/9–2018/6
· 西南交通大学 | 四川峨嵋 | 汉语水平考试四级

Bachelor of Science: Mechanical Engineering 2014/9–2017/6
· University of Applied Sciences Upper Austria | Austria

Projects

Robomaster 2021/10–2022/4
SOFTWARE TEAM · HKUST
- SLAM and navigation for autonomous Robots in ROS
- Embedded software development on STM32.

Autonomous RC-car race (first place 🏆) 2020/12–2021/3
IMITATION LEARNING · HKUST https://github.com/JosefGst/autorace
- Trained Pytorch model to be used on Jetson Nano for autonomous-driving, obstacle avoidance and overtaking of other cars.